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FROM A NONLINEAR KINETIC EQUATION TO A VOLUME-EXCLUSION CHEMOTAXIS MODEL VIA ASYMPTOTIC PRESERVING METHODS

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Abstract. In this work we first prove, by formal arguments, that the diffusion limit of nonlinear kinetic equations, where both the *transport term and the turning operator are density-dependent*, leads to volume-exclusion chemotactic equations. We then numerically study this diffusive limit via an asymptotic preserving scheme based on a micro-macro decomposition. By properly discretizing the nonlinear term implicitly-explicitly in an upwind manner, the scheme produces accurate approximations also in the case of strong chemosensitivity. We show, via detailed calculations, that the scheme is asymptotic preserving and bound preserving and show numerically an energy dissipation property, which are essential for practical applications. We extend this scheme to two dimensional kinetic models and we validate its efficiency by means of 1D and 2D numerical experiments of pattern formation in biological systems.

1. **Introduction.** Chemotaxis is the mechanism by which cells and organisms adapt their movement in response to a chemical stimulus present in their environment. This phenomenon has been observed in many biological systems [**?**, **?**, **?**, **?**].

The mathematical study of chemotaxis started from the seminal contributions of Patlak [**?**], Keller, Segel and Alt [**?**, **?**, **?**], where the authors introduced the celebrated Patlak-Keller-Segel-Alt (PKSA) model. This model was originally proposed for pattern formation in bacterial populations through an advection-diffusion system of two coupled parabolic equations describing the evolution of the cell density and the chemoattractant. The PKSA model has become the prevailing method for representing chemotactic behaviour in biological systems at population level (see Ref. [**?**] for a review about Keller-Segel type models). A drawback of studying biological systems directly at the continuous level, i.e. through partial differential equations (PDEs), is that the interactions comprised in such models can be difficult to interpret in terms of mechanisms at the cellular level. Therefore it is fundamental to explain the origin of these macroscopic models from a mechanistic/microscopic description of motion. In the case of the PKSA model, tremendous amount of works in this direction have been made by various authors (see reviews

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of Horstmann [?, ?]). Among other approaches, several authors showed that the chemotaxis models could be obtained from biased random walk approaches [?, ?], or as the parabolic limit of velocity-jump processes / transport models [?, ?, ?]. For instance in [?], the authors study the diffusion-limit of transport equations under a variety of external biases imposed on the turning operator. Depending on the strength of the bias, the authors show that this leads to anisotropic diffusion, drift term in the flux, or both. In particular, they show that the PKSA model only arises if the perturbation of the turning kernel is sufficiently small. They consider several examples of prototype models for chemotaxis of bacteria, of slime molds and of myxobacteria and present a general discussion of the derivation of diffusion approximations from various stochastic processes that model chemotaxis (see [?] and references therein). These works not only give insights into how the continuum PKSA model relates to models at the microscopic scale, but also enable to understand the link between different microscopic models and modeling assumptions.

The PKSA model has also been extensively studied in the literature from a theoretical viewpoint. Of particular interest is the tendency of solutions to exhibit finite-time blow-up [?, ?, ?, ?, ?]. It is now well established that in dimension $n \geq 2$, finite-time blow-up may occur if the initial condition exceeds some threshold. However, under the biologically relevant cases for aggregation to occur, initial conditions typically lie above this threshold. For these cases, the solutions of the PKSA model would correctly predict pattern formation but in the form of blow-up, i.e. the model will not be relevant for later stages following aggregation in real systems. Therefore, it is necessary to modify the model to allow pattern formation without blow-up. In this direction, the PKSA original model has been modified by various authors with the aim of improving its consistency with biological systems (see [?] for a comprehensive review). One example is the volume-exclusion chemotactic system (VEPKS) introduced by Hillen and Painter [?] to take into account the finite size of cells and volume limitations. In such models the chemotactic sensitivity (i.e. the term leading to cell aggregation) depends on both the chemical concentration in the medium and the local cell density, thus, the population directly modulates its own sensitivity response. The coupled system reads, in its parabolic-elliptic form as

$$\begin{aligned} \partial_t \rho - \nabla \cdot \Big(D_0 (q(\rho) - \rho q'(\rho)) \nabla \rho - \chi_0 q(\rho) \rho \nabla c \Big) = h(\rho, c) \ , \quad t \geq 0 \ , \mathbf{x} \in \Omega \subset \mathbb{R}^n \ , \\ D_c \Delta c + g(\rho, c) = 0 \ . \end{aligned} \tag{1}$$

Here, $\rho(t, \mathbf{x})$ is the cell density, $c(t, \mathbf{x})$ is the chemoattractant concentration and $q(\rho)$ is a function that describes the packing capacity of the cell aggregates. The diffusion coefficients for the cells and chemoattractant are D_0 and D_c, respectively, and χ_0 is the chemotactic sensitivity. The proliferation (or death) of the cells is described by $h(\rho, c)$, and the production and consumption of the chemoattractant is given by $g(\rho, c)$. The dependency of h and g on the cell density ρ enables to account for density-limited phenomena such as the logistic-growth function for h commonly used in the literature [?, ?]. Note that from (1), we recover the classical PKSA model by taking $q(\rho) = 1$. It has been shown [?, ?, ?] that such volume-exclusion effects prevent blow-ups in finite time compared to the model without density effects (with $q(\rho) = 1$). The volume-exclusion chemotactic equation has been widely studied in the literature, from a modeling [?, ?], analytic [?, ?, ?, ?]

and numerical perspectives [?]. A relevant example where volume-exclusion effects play a crucial role is in the modeling of glioblastoma (GBM) aggregates [?]. In particular, the system (1) was used in [?] to explain the mechanical changes at the cell level due to the presence of a chemical treatment. In [?], the cell's elasticity is modeled through the term $q(\rho)$ which incorporates the cell-cell interactions [?]. This function $q(\rho)$ was chosen to be $q(\rho) = 1 - (\rho/\bar{\rho})^\gamma$ where γ is a parameter that depends on the concentration of the treatment and $\bar{\rho}$ is the maximum cell density in each aggregate. For $\gamma = 1$ the cells were considered as solid particles while $\gamma > 1$ corresponded to semi-elastic particles that can squeeze into empty spaces.

In one dimension, Hillen and Painter [?] derived the VEPKS from a discrete-space random walk. The derivation begins with a master equation for a continuous-time and discrete space random walk model as in [?], but including the volume-filling effect by making the probability of making a jump depend on the availability of space into which cells can move. A natural question that arises is whether the VEPKS model can be obtained as the limit of a kinetic velocity-jump model as done for the PKSA model. In this paper, our first aim is to provide an interpretation of the volume-exclusion chemotactic system (1) as the diffusion limit of a kinetic velocity-jump model.

Inspired by the approach in [?, ?], we first show, by formal arguments, that the system (1) can be obtained in the diffusion limit of a nonlinear kinetic equation, provided that both the transport term and the turning operator are density-dependent (Theorem ?? in Section ??). The corresponding kinetic velocity-jump model we propose reads

$$\partial_t f + \mathbf{v} \cdot \nabla\big(F[\rho](t,\mathbf{x},\mathbf{v}) f\big) = \lambda q(\rho)\big(- f + \rho T(\mathbf{v},\rho,\nabla c)\big) + r_0 f \Big(1 - \frac{\rho}{\rho_{\max}}\Big)_+ , \quad (2)$$

where $f(t,\mathbf{x},\mathbf{v}) \geq 0$ is the phase space cell density, $\mathbf{x} \in \Omega \subset \mathbb{R}^n$ denotes the position, $\mathbf{v} \in V \subset \mathbb{R}^n$ is the velocity where V denotes the unit sphere $V = \{\mathbf{x} \in \mathbb{R}^n : \ |\mathbf{x}| = 1\}$ and $t \in \mathbb{R}^+$ the time. Note that we have considered that cells proliferate according to a logistic-growth law $h(\rho,c) \equiv h(\rho) = r_0 f\Big(1 - \frac{\rho}{\rho_{\max}}\Big)_+$ where $\rho_{\max}$ is the carrying capacity and r_0 is the proliferation rate. In (2), λ is the constant turning rate, with $1/\lambda$ giving a measure of the mean run length between velocity jumps, $T(\mathbf{v},\rho,\nabla c)$ gives the probability of a velocity jump to velocity $\mathbf{v}$, which accounts for cell preference to move up the chemotactic gradient ∇c, and the term $F[\rho](t,\mathbf{x},\mathbf{v})$ describes the anisotropic transport due to the density limited motion. The fact that the turning operator $T(\mathbf{v},\rho,c,\nabla c)$ depends on the gradient of chemical concentration ∇c is necessary to obtain the chemotactic behavior described by (2), as discussed thoroughly in [?]. For simplicity, we have assumed here that the turning rate only depends on ∇c but the derivation extends to more general choices $T(\mathbf{v},\rho,c,\nabla c)$, as briefly discussed in remark ??. While similar models have been considered in the literature, [?, ?], ours extends to higher dimensions and includes an extra nonlinearity in the transport term.

In model (2), it is noteworthy that cell motion is not only biased by external factors (leading to chemotactic motion) but also biased to account for density-limited motion/crowding effects. Model (2) shows that the minimal hypothesis on the run-and-tumble process enabling to obtain the VEPKS system (1) in the diffusion limit are density-dependent transport term and turning kernel. The first hypothesis (density-dependent transport) is the most natural in terms of a modeling viewpoint, and expresses that cells will only be able to move in not already crowded regions. The second hypothesis (density-dependent turning kernel) is a more 'active' mechanism expressing that cells have less probability to turn towards crowding regions. Applied to bacteria, these hypothesis are consistent with experimental evidence showing that tightly packed colonies could collectively move faster by reducing the speed of their individual bacteria [**?**].

The second goal in this paper is to numerically investigate whether the relevant macroscopic volume exclusion equation corresponds to the underlying physical system described by the kinetic equation in the diffusion limit. These methods are widely known as asymptotic preserving schemes (AP) [**?**, **?**] since they mimic the asymptotic behaviour of the kinetic equation when the scaling parameter approaches to zero and the mesh size and time steps are fixed. In this paper, we use a micro-macro decomposition of the unknown in the sense of [**?**] as detailed in Section **??**. The finite difference discretization is explained in Section **??**, where a proper implicit-explicit discretization of the nonlinear terms, including the chemosensitivity term, is applied to improve the efficiency and stability of our numerical scheme. This decomposition of the solution of the kinetic equation is analogous to a Chapman-Enskog expansion in the case of the classical Boltzmann equation. It uses the properties of the "collisional operator", which in our formulation describes the run and tumble movement of the individual. For the kinetic counterpart of the classical PKSA equation we refer to [**?**], where the authors used an odd-even splitting at the kinetic level and studied the blow-up behaviour of solutions. Other related works can be found in [**?**, **?**, **?**, **?**].
Outline of the paper. This paper is organized as follows. In Section **??** we introduce the new version of the kinetic velocity-jump model given by (2) and consider its diffusion limit using a Hilbert expansion method as in Ref. [**?**] under appropriate scaling assumptions for the turning operator. We show that the limiting system has a dissipating free energy and we derive an energy estimate used to validate the numerical method. In Section **??**, we present the micro-macro decomposition of the kinetic model that we use to build the AP scheme presented, and analyzed various properties (positivity preserving, AP etc) in Section **??**. Finally, we present the numerical experiments and draw conclusions in Section **??**.

2. Examples.

2.1. A sample Theorem.

Theorem 2.1. *Content of your theorem.*

Proof. To refer to equations in your article, use the commands: (3), (4) and (6). □

2.2. A sample Lemma.

Lemma 2.2. *Content of your lemma.*

Proof. Your proof statements. □

2.3. A sample Remark.

Remark 2.3. Content of your remark.

2.4. **A sample Definition.**

Definition 2.4. Sample: Let ϕ_t be an Anosmia flow on a compact space V and $A \subset V$ a dense set. Say that the upper Lacunae exponents are $\frac{1}{2}$-*pinched* on A if

$$\sup_{x \in A} \frac{\max\{|\bar{\lambda}| : \bar{\lambda} \text{ is a nonzero upper Lyapunov exponent at } x\}}{\min\{|\bar{\lambda}| : \bar{\lambda} \text{ is a nonzero upper Lyapunov exponent at } x\}} \leq 2. \tag{3}$$

2.5. **A sample Proposition.**

Proposition 2.5. *Content of your proposition.*

2.6. **A sample Corollary.**

Corollary 2.6. *Content of your corollary.*

2.7. **A sample Assumption.**

Assumption 2.7. Content of your assumption.

2.8. **Example of inserting a Figure.** Make sure all figures and tables fit within the margins.

FIGURE 1. Here is the Caption of your figure

3. **How to align the math formulas.**

Theorem 3.1. *Content of your theorem.*

In the proof below, we will to show you how to align the math formulas:

Proof of Theorem 3.1. Please refer to the following example to align your math formulas:

$$\begin{aligned} \theta_\varepsilon \wedge d\theta_\varepsilon^{n-1} &= (\theta_0 + \varepsilon\alpha) \wedge (d(\theta_0 + \varepsilon\alpha))^{n-1} \quad \text{since } d\alpha = 0 \\ &= (\theta_0 + \varepsilon\alpha) \wedge (d\theta_0)^{n-1} + \theta_0 \wedge d\theta_0^{n-1} - \varepsilon d(\alpha \wedge \theta_0 \wedge d\theta_0^{n-2}) \\ &\quad + \theta_0 \wedge d\theta_0^{n-1} + \varepsilon\alpha \wedge d\theta_0^{n-1} \\ &= \theta_0 \wedge d\theta_0^{n-1} - \varepsilon d(\alpha \wedge \theta_0 \wedge d\theta_0^{n-2}), \end{aligned} \tag{4}$$

It can also be aligned in the following way:

$$\begin{aligned}
&\theta_\varepsilon \wedge d\theta_\varepsilon^{n-1} \\
&= (\theta_0 + \varepsilon\alpha) \wedge (d(\theta_0 + \varepsilon\alpha))^{n-1} \quad \text{since } d\alpha = 0 \\
&= (\theta_0 + \varepsilon\alpha) \wedge (d\theta_0)^{n-1} + \theta_0 \wedge d\theta_0^{n-1} - \varepsilon d(\alpha \wedge \theta_0 \wedge d\theta_0^{n-2}) \\
&\quad + \theta_0 \wedge d\theta_0^{n-1} + \varepsilon\alpha \wedge d\theta_0^{n-1} \\
&= \theta_0 \wedge d\theta_0^{n-1} - \varepsilon d(\alpha \wedge \theta_0 \wedge d\theta_0^{n-2}),
\end{aligned} \tag{5}$$

Here is another example for if the math expression in [ ] must be split to a new line:

$$\begin{aligned}
\int_0^T |u_0(t)|^2 dt \leq \delta^{-1} \Bigg[&\int_0^T (\beta(t) + \gamma(t)) dt \\
&+ T^{\frac{2(p-1)}{p}} \left(\int_0^T |\dot{u}_0(t)|^p dt \right)^{\frac{2}{p}} + T^{\frac{2(p-1)}{p}} \left(\int_0^T |\dot{u}_0(t)|^p dt \right)^{\frac{2}{p}} \Bigg].
\end{aligned} \tag{6}$$

Please use displaystyle if your formulas fully occupy a paragraph and use textstyle for formulas among text.

For two equations:

$$A = \theta_0 \wedge d\theta_0^{n-1} - \varepsilon d(\alpha \wedge \theta_0 \wedge d\theta_0^{n-2})$$
$$B = \theta_1 \wedge d\theta_1^{n-1} - \varepsilon d(\alpha \wedge \theta_1 \wedge d\theta_1^{n-2})$$

Please align your formulas nicely according to the above examples. Thank you. □

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REFERENCES

[1] F. Abergel and R. Tachet, work in progress.

[2] S. Arora, M. T. Mohan and J. Dabas, Approximate controllability of the non-autonomous impulsive evolution equation with state-dependent delay in Banach space, to appear, *Nonlinear Anal. Hybrid System.*

[3] S. Ball, Polynomials in finite geometries, *Surveys in Combinatorics, 1999 (Canterbury)*, London Math. Soc. Lecture Note Ser., 267, Cambridge University Press, Cambridge, 1999, 17-35.

[4] Y. Benoist, P. Foulon and F. Labourie, Flots d'Anosov a distributions stable et instable differentiables, *J. Amer. Math. Soc.*, **5** (1992), 33-74.

[5] T. Pang and A. Hussain, An application of functional Ito's formula to stochastic portfolio optimization with bounded memory, *Proceedings of the SIAM Conference on Control and Its Applications*, Paris, France, 2015, 159-166.

[6] *SARS Expert Committee, SARS in Hong Kong: From Experience to Action*, Report of Hong Kong SARS Expert Committee, 2003. Available from: http://www.sars-expertcom.gov.hk/english/reports/reports.html.

[7] J. Serrin, Gradient estimates for solutions of nonlinear elliptic and parabolic equations, in *Contributions to Nonlinear Functional Analysis*, Academic Press, 1971, 33-75.

[8] J. Smoller, *Shock Waves and Reaction-Diffusion Equations*, 2^{nd} edition, Springer-Verlag, New York, 1994.

[9] A. Teplinsky, Herman's theory revisited, preprint, 2012, arXiv:0707.0078.

[10] K. Wei, *Torsion Cycles and Set Theoretic Complete Intersection*, Ph.D thesis, Washington University in St. Louis, 2006.

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